

ALOK RANJAN

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PROFESSIONAL SUMMARY

M.Tech student in Aerospace Engineering (Propulsion) at IIT Kanpur with expertise in Computational Fluid Dynamics, parallel computing, and numerical methods. Experienced in custom OpenFOAM solver development, HPC workflows, and FE simulations. Adept at managing end-to-end CFD pipelines from geometry preparation to post-processing.

EDUCATION

M.Tech, Aerospace Engineering (Propulsion) IIT Kanpur <i>Qualified GATE 2024 (Aerospace Engineering)</i>	2024 – Present
B.Tech, Aerospace Engineering IIST Thiruvananthapuram <i>Selected through JEE Advanced 2018</i>	2018 – 2022
Senior Secondary (XII) – Prayas Boys Residential School, Raipur 	2017
Secondary (X) – Loyola Higher Secondary School, Kunkuri 	2015

RESEARCH EXPERIENCE & PROJECTS

M.Tech Thesis: Numerical Investigation of Electrostatic Spray Dynamics | IIT Kanpur Jul 2025 – Present
Guide: Dr. Ashoke De, IIT Kanpur

- Developed sprayElectroFoam — a custom OpenFOAM solver with EHD coupling, Lagrangian tracking, and image-charge forces.
- Implemented RANS and LES turbulence models on HPC to analyze transient spray plume dynamics and droplet stability.
- Post-processed high-density simulation datasets; validated droplet morphology against experimental benchmarks using Python.
- Managed end-to-end CFD workflows: geometry, meshing, parallel solver execution, and statistical data extraction.

B.Tech Thesis: Failure Analysis of Bio-Inspired Hierarchical Composites | IIST Thiruvananthapuram Jan – May 2022
Guide: Dr. Anup S, IIST

- Performed FE simulations in ABAQUS to study strain fields and stress distribution in hierarchical composites.
- Designed and fabricated 1H and 2H hierarchical composites via 3D printing; conducted and analyzed tensile tests.

Internship: CFD Analysis of 2D Flow past Square Cylinder | IIST Thiruvananthapuram Jun – Oct 2021
Guide: Dr. Manu K V

- Simulated unsteady flow and vortex shedding at low Reynolds numbers using OpenFOAM.
- Computed lift/drag coefficients and Strouhal numbers; validated against published benchmark data.

Course Project: Aerospace Vehicle Design | IIST / VSSC, ISRO Aug – Nov 2021
Mentors: Dr. Ayyappan G (VSSC, ISRO) & Dr. Manoj T Nair (IIST, ISRO)

- Designed and optimized IAD geometry and parachute systems; developed trajectory analysis codes.

Course Project: Parallel Computation for Sparse Matrix Systems | IIT Kanpur Jan – May 2025
Instructors: Dr. Ashoke De & Dr. Malay K Das

- Implemented serial, OpenMP, and MPI solvers; benchmarked performance and scalability using PETSc.

TECHNICAL SKILLS

CFD & Simulation: OpenFOAM, ABAQUS, ICEM CFD, Gmsh, ParaView, SolidWorks, CATIA, Autodesk Inventor

Programming & HPC: Python, C++ (OpenFOAM), OpenMP, MPI, PETSc, HPC cluster workflows

Numerical Methods: Numerical Linear Algebra, Iterative Solvers, RANS, LES, Lagrangian Tracking, EHD coupling

Documentation: LaTeX, Microsoft Office

RELEVANT COURSEWORK

M.Tech: Parallel Computation, Numerical Linear Algebra, Computational Methods, Fluid Dynamics, Combustion, Heat Transfer, Compressible Flow

B.Tech: Air-Breathing Propulsion, Thermodynamics, Fluid Mechanics, Heat Transfer, Optimization Techniques in Engineering

ACHIEVEMENTS

- Qualified GATE 2024 (Aerospace Engineering) — Selected for M.Tech Propulsion at IIT Kanpur.
- Selected for B.Tech Aerospace at IIST Thiruvananthapuram through JEE Advanced 2018.